**Table 1**

| Design/Control Parameters for the Terrain Adaptive control | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Incline Gradient (θ)** | **Terrain State** | **Normal Budget (Floadcosθ)** | **Adaptive Stiffness (k)** | **Nominal Damping (c)** | **Target Static Fground** | **Required Traction (μreq)** | **Traction SF (μ=0.60)** |
| 0∘ | Level Ground | 392.4 N | 2182 N/m | 155.0 N·s/m | 285.8 N | 0 | ∞ |
| 10∘ | Gentle Slope | 386.4 N | 2149 N/m | 150.0 N·s/m | 279.8 N | 0.176 | 3.4 |
| 20∘ (Nominal) | Design Slope | 368.7 N | 2050 N/m | 145.0 N·s/m | 269.3 N | 0.364 | 1.65 |
| 25∘ | Steep Incline | 355.6 N | 1977 N/m | 138.0 N·s/m | 259.0 N | 0.466 | 1.29 |
| 30∘ | Critical Limit | 339.8 N | 1890 N/m | 130.0 N·s/m | 246.5 N | 0.577 | 1.04 |

| Tuned Gain Parameters for 20 degrees mountainous incline | | | | |
|---|---|---|---|---|
| **Parameter** | **Symbol** | **Value** | **Unit** | **Functional Role** |
| Proportional Gain | Kp | 320 | N·m/rad | Primary corrective stiffness against frame pitch |
| Integral Gain | Ki | 45 | N·m/(rad·s) | Cancels steady-state sag on sustained 20∘ slopes |
| Derivative Gain | Kd | 28.5 | N·m·s/rad | Suppresses resonance and overshoot at heel-strike |
| Sampling Frequency | fs | 100 | Hz | Sensor acquisition and control update rate |
| Stroke Extension Limit | ΔL5 | ±0.125 | m | Telescopic actuator stroke saturation bounds |

**Table 2**: Summary of material properties and constituent assumptions[1]

| Component | Material | Primary function | Constituent Model | Required Properties |
|---|---|---|---|---|
| Main frame | Carbon fibre (CFRP) | Structural integrity & weight minimization | Linear-elastic orthotropic carbon/epoxy lamina, assembled as a symmetric quasi-isotropic laminate. Material coordinate system follows the principal fibre direction of the frame. | $E_1$ = 135 GPa; $E_2 = E_3$ = 10 GPa;<br>$V_{12}$ = 0.30; $V_{13}$ = 0.3; $V_{23}$ = 0.4;<br>$G_{12}$ = 5.0; $G_{13}$ = 5.0; $G_{23}$ = 3.8<br>P = 1600 $kgm^{-2}$;<br>$X_t$ = 1500 Mpa; $X_C$ = 1200 Mpa;<br>$Y_t$ = 50 Mpa; $Y_C$ = 200 Mpa; $S_{12}$ = 70 Mpa<br>Laminate = $[0/+45/-45/90]_{st}$ with 0.25mm ply thickness; frame thickness = 5mm uniform<br>Frame wt. original = 5.1kg, Optimized = 4.1kg |
| Active Suspension Hinge (ASH) | CFRP/ Ti-6AI-4V | Impact force attenuation | Isotropic linear-elastic Titanium alloy solid; non-linear contact at hinge interfaces if relative articulation is represented. | E = 114 GPa; V = 0.34; ρ = 4430 $kgm^{-3}$;<br>Yield strength $\sigma_y$ = 880 Mpa<br>Ultimate tensile strength $\sigma_u$ = 950Mpa<br>From 2.5mm to 6mm. |
| Spring Damper | Ti-6AI-4V | Impact force attenuation | Isotropic Titanium structural solid combined with a spring damper connector model | E = 114 GPa; V = 0.34; ρ = 4430 $kgm^{-3}$;<br>$\sigma_y$ = 880 Mpa;<br>C ; K = |
| Telescopic support rods | Grade 5 Ti / Al Alloy | Load by-pass to the ground | Telescoping structural members with contact between the internal & external tube sections. | Ti-6AI-4V: E = 114GPa; V = 0.34; ρ = 4430$kgm^{-3}$; $\sigma_y$ = 880Mpa;<br>Al alloy (7075-T6): E = 71.7 GPa; V = 0.33; ρ = 2810 $kgm^{-3}$; $\sigma_y$ = 503 Mpa; |
| Omni-Wheels (OMDW) | CFRP rim/wheel body | Multi-axis mobility on narrow paths | Linear, elastic orthotropic carbon/epoxy laminate. Local material axes follows the principal fibre directions around the wheel structure. | $E_1$ = 135 GPa; $E_2$ = 10 GPa; $E_3$ = 10 GPa;<br>$V_{12}$ = 0.30; $V_{13}$ = 0.3; $V_{23}$ = 0.4;<br>$G_{12}$ = 5.0; $G_{13}$ = 5.0; $G_{23}$ = 3.8<br>P = 1600 $kgm^{-2}$;<br>$X_t$ = 1500 Mpa; $X_C$ = 1200 Mpa;<br>$Y_t$ = 50 Mpa; $Y_C$ = 200 Mpa; $S_{12}$ = 70 Mpa<br>Laminate = $[0/+45/-45/90]_{st}$ with 0.25mm ply thickness |
| | Titanium Hub & Axle | | Isotropic, linear-elastic structural solid | E = 114GPa; V = 0.34; ρ = 4430 $kgm^{-3}$; $\sigma_y$ = 880Mpa; $\sigma_u$ = 950 Mpa |
| | Polyurethane roller tread | | Isotropic, hyper-elastic or simplified linear elastic compliant layer. | Linear elastic approx.<br>E = 25 Mpa; V = 0.49; ; ρ = 1100 $kgm^{-3}$<br>Nominal thread harness: Shore A 85 – 95 |
| | Wheel ground interface | | Nonlinear frictional contact or compression only support simplified static load transfer. | Dry rock friction coefficient (μ) = 0.6;<br>Sensitivity range = $0.4 \leq \mu \leq 0.70$ |
| | Rolling resistance | | Equivalent resisting moment or force opposing wheel rotation & translation | Coefficient of resistance:<br>- For firm, dry terrain, $C_{RR}$ = 0.03<br>- Rough, loose or rocky terrain, $C_{RR}$ = 0.05 – 0.10<br>Resistive force ($F_{RR}$) = $C_{RR}N$ |
| Ergonomic mesh harness | Antimicrobial Nylon mesh/ Velcro | User comfort & heat dissipation | Flexible orthotropic fabric or simplified compliant load-transfer interface. | $G_{Warp}$ = 0.5 GPa; $E_{Weft}$ = 0.35 GPa; V = 0.39;<br>ρ = 1150 $kgm^{-3}$; Tensile strength = 70 Mpa;<br>nominal thickness = 2.0 mm |

*Source: FreeCAD/Authors*

[1] The structural analysis assumes linear elastic behaviour within the range of the applied loads. No claim of progressive composite damage, delamination or post-failure behaviour is made in the present analysis.

Table 3: Anthropometric and biomechanical properties of the virtual human model

| Parameter | Value | Source/Justification |
|---|---|---|
| Sex | Female | Study design |
| Anthropometric percentile | $50^{th}$ | Study design |
| Body mass | 75kg | Prescribed simulation condition |
| Stature | 1.65 m | Representative modelling estimate |
| Head & neck mass | 5.01 kg | 6.68% of body mass [47] |
| Trunk mass | 31.93 kg | 42.57 of body mass [47] |
| Upper arm mass | 1.91 kg each | 2.55% per segment [47] |
| Forearm mass | 1.04 kg each | 1.38% per segment [47] |
| Hand mass | 0.42 kg each | 0.56% per segment [47] |
| Thigh mass | 11.09 kg each | 14.78% per segment [47] |
| Shank mass | 3.61 kg each | 4.81% per segment [47] |
| Foot mass | 0.97 kg each | 1.29% per segment [47] |
| Head & neck segment length | ≈ 0.21 m | Representative anthropometric estimate |
| Trunk length | ≈ 0.52 m | Representative anthropometric estimate |
| Upper arm length | ≈ 0.305 m each | Representative anthropometric estimate |
| Forearm length | ≈ 0.240 m each | Representative anthropometric estimate |
| Hand length | ≈ 0.175 m each | Representative anthropometric estimate |
| Thigh length | ≈ 0.405 m each | Representative anthropometric estimate |
| Shank length | ≈ 0.400 m each | Representative anthropometric estimate |
| Foot length | ≈ 0.250 m each | Representative anthropometric estimate |
| Cervical/head joint | 3 rotational DOF | Head-truck articulation |
| L5/S1 joint | 3 rotational DOF | Truck-pelvis articulation & spinal-load assessment |
| Hip joints | 3 rotational DOF each | Spherical hip representation |
| Knee joints | 1 primary rotational DOF each | Each Flexion/extension |
| Ankle joints | 1 primary rotational DOF each | Each Dorsiflexion/plantar flexion |
| Shoulder joints | 3 rotational DOF each | Each spherical shoulder representation |
| Elbow joints | 1 primary rotational DOF each | Each Flexion/extension |
| Writs joints | 2 rotational DOF each | Each Flexion/extension & radial/inner deviation |
| Walking speed | 1.2 m/s | Simulation condition |
| Terrain inclination | 20 deg. | Simulation condition |
| Payload | 40kg (≈ 400 N) | Design-standard condition |
| Load carriage distance | 4.3 Km | Field data for LCA operational estimation |
| Trips per day | 2.77 | Field data for LCA operational estimation |
| Assessment period | 5 years | Field data for LCA operational estimation |
| Terrain coefficient ($\eta$) | 1.2 | Simulation condition |

***Note:*** segment mass and lengths were computed using the female segment masses fractions reported by de leva [47]. This provides adjusted anthropometric segment parameters referenced to commonly as anatomical joint centres.

For each body segment; $M_i = f_i M$

Where

$M_i$ = Mass of segment I,
$f_i$ = female segment mass fraction;
$M$ = total body mass (75 kg)

Table 4: Summary of principal MBD connections

| Human-Device Component | MBD Connections | Mechanical function |
|---|---|---|
| Shoulder/Lifter straps - human/frame | Fixed attachment | Transfers load between the harness & frame |
| Lumbar/Hip belt – human/frame | Fixed attachment | Maintains frame-body alignment & transfers interface forces |
| CFRP frame | Rigid body | Provides structural backbone of the device |
| CFRP frame – Active suspension hinge | Revolute joint | Permits controlled angular motion about the hinge axis |
| Hinge to Telescopic support assembly | Rigid/Revolute connection | Transfers suspension motion & force to support system |
| Telescopic rod | Translational/Prismatic joint | Permits axial extension & compression |
| Spring damper - Telescopic rod | Compliant force element | Represents elastic & damping response |
| Telescopic rod - wheel assembly | Rigid/Appropriate joint | Transfers support forces to wheel |
| Wheel – Wheel axle | Revolute joint | Allows wheel rotation |
| Wheel - Terrain | Contact/Friction interaction | Transfers load to the ground & governs traction/slip |

Table 5

| **Payload Coordinates relative to Stance Foot contact - A** | | | | | |
|---|---|---|---|---|---|
| **Link (i)** | **Segment Description** | **Length $L_i$ (m)** | **Angle ($\varphi_i$) Degrees** | **$\Delta X_i = L_i Cos\varphi_i$ (m)** | **$\Delta Y = LiSin\varphi i$ (m)** |
| 1 | Shank | 0.400 | 75.0 | +0.1035 | +0.3864 |
| 2 | Thigh | 0.405 | 65.0 | +0.1712 | +0..3671 |
| 3 | Trunk (Hip – Mount) | 0.520 | 80.0 | +0.0903 | +0.5121 |
| 4 | Frame Stand-off) | 0.240 | 170.0 | -0.2364 | +0.0417 |
| 5 | Trailing Strut ($L_5$) | 0.900 | 215.0 | -0.7372 | -0.5162 |

**Payload position relative to OMDW contact**

Horizontal distance to rear wheel ($X_L$) = -0.6086 m (60.86 cm)
Vertical height above wheel ($Y_L$) = +0.7911 m (79.11 cm)
Resultant payload vector magnitude = PL = 0.9981 m

| **Summary Virtual Human Body Mass inputs - B** | | | |
|---|---|---|---|
| **Segment** | **Mass per Unit (Kg)** | **Quantity** | **Total Sub-mass (Kg)** |
| Head/Neck | 5.01 | 1 | 5.01 |
| Trunk | 31.93 | 1 | 31.93 |
| Upper arms | 1.91 | 2 | 3.92 |
| Forearms | 1.04 | 2 | 2.08 |
| Hand | 0.42 | 2 | 0.84 |
| Thigh | 11.09 | 2 | 22.18 |
| Shank | 3.61 | 2 | 7.22 |
| Foot | 0.97 | 2 | 1.94 |
| **Total Virtual Human Body Mass** | - | - | **75.12** |

Payload weight ($W_P$) = 40 kg * 9.81 m/s$^2$ = 400 N
Human Body Weight ($W_{body}$) = 75 * 9.81) = 735.9 N
Total Combined Gravity Load ($W_{Total}$) = 1135.75 N (1128.34 – 1135.75)

| | **Summary of L5/S1 and Hip dynamic joint burden - C** | | |
|---|---|---|---|
| **Segment** | **Mass per Unit (Kg)** ***(refer Annexure A)*** | **Quantity** | **Total Sub-mass (Kg)** |
| Head/Neck | 5.01 | 1 | 5.01 |
| Trunk | 31.93 | 1 | 31.93 |
| Upper arms | 1.91 | 2 | 3.92 |
| Forearms | 1.04 | 2 | 2.08 |
| Hand | 0.42 | 2 | 0.84 |
| **Total Upper body mass ($M_{Upper}$)** | - | - | **43.68** |

Net compressive force across L5/S1 Joint with LoCaDe-Exo bypass *($F_{L5/S1}$)* = 548.5 N

Total compressive force ($F_{Unassisted}$)

Because there is no bypass mechanism in the unassisted condition, 100% of the payload weight *($W_{payload}$)* is borne directly by the torso/spine

Therefore, $F_{Unassisted}$) = $W_{Upper\ body}$ + $W_{payload}$ = 428.50 N + 400 N = 828.50 N

**Summary of force transmission model outputs - D**

| Variable | Governing formulae | Static value | Peak transient value | Unit |
|---|---|---|---|---|
| Normal gravitational term | $F_{Load}$*Cosθ | 375.88 | 568.38 | N |
| Active Suspension Hinge resistance ($F_{damp}$) | $C\dot{x} + Kx$ | 106.00 | 87+ 61 = 148.50 | N |
| Ground normal force ($F_{ground}$) | Fload Cosθ - Fdamp | 269.28 | 418.88 | N |
| Ground shunt percentage (%) | (Fground/Fload)*100 | 71.68 | 73.8 | % |

$F_{Load}$ = 40 kg * 9.9812 = 392.4 N (or 400N nominal)

$F_{Load}$ * Cos θ = 392.4 Cos $20^0$ = 368.7 N

**Global Force Equilibrium Formulation (GFE) - E**

Payload weight ($W_P$) = 40 kg * 9.81 m/s$^2$ = 400 N

Total Combined Gravity Load ($W_{Total}$) = 1135.75 N

Ground Shunt percentage = 71.68%

| | | |
|---|---|---|
| Ground Shunt proportion | $N_{Shunt}$ | 286.72 N |
| Physiological retention proportion | $N_{Ret.}$ | 113.48 N |

**Ground Reaction Force at Omni-wheel (OMDW) Contact - F**

| | | | |
|---|---|---|---|
| Ground vertical GRF | *Fy, wheel* | 286.52 | N |
| Terrain normal force | *Fn* | 269.24 | N |
| Terrain-Longitudinal Friction/Downslope force | $F_{II}$ | 97.98 | N |
| Terrain margin verification | $U_{req.} = F_{II} / Fn$ = tan ($20^0$) | 0.364 | |
| Payload offset distance | $d_{p,H}$ | 0.22 | M |
| Effective moment arm | $L_{Movement}$ | 0.733 | M |
| F*strut* | $W_P * d_{p,H} / L_{Movement}$ | 120.5 nominal | N |
| F*human* | F*human* = F*strut* | ≈ 120 | N |
| Fground | Wp - F*strut* | ≈ 280 | N |

$U_{req.} \approx$ 0.364 ≤ U*dry,rock (0.60)*

Safety Factor against wheel slip (SF) = U*terrain*/U*req*. = 0.60/0.360 = 1.65 *(No Slip)*

$$\sum M_H = \mathbf{0} \implies W_P * d_{p,H} - F_{Strut} * L_{Movement} = 0 \implies 88 - 120 * 0.733 = 0$$

Where,

$d_{p,H}$ = payload centre of mass offset behind Hip pivot

$L_{Movement}$ = perpendicular lever arm of the trailing strut

**Table 6:**

| Field Data from Dunda Block, Uttarkashi district, Uttarakhand state | | | | |
|---|---|---|---|---|
| **Variable** | Distance covered during load carriage? | How many trips do you need to make to complete a task? | On average, how much weight do you carry in a single load? (kg) | How many days do you carry loads in a month? |
| **Mean** | 4.3836 | **2.77** | ≈ 40 | 26 |
| **SD** | **3.99** | **0.8604** | 12.55 | |
| **Pandolf-Santee Model** | | | | |
| $D_{month}$ | 316.3 Km/month | | | |
| $E_{Life,i}$ | 18,980 Km | | | |
| Human mass (W) | 75 Kg | | | |
| Payload (L) | 40 Kg | | | |
| Walking speed | 1.2m/s | | | |
| Inclination | 20 deg. | | | |
| Grade (G) | 36.40% | | | |
| $EF_{food}$ | = 0.20 Kg $CO_2e$/MJ | | | |
| Terrain coefficient (η) | **1.2** | | | |

**Table 7: Modal analysis parameters**

| | |
|---|---|
| • Cadence (step frequency) = F =V/Lstep<br>• Typical walking level-ground step length = 0.55 – 0.7 m<br>• Adjustment for 20 deg. Incline 0.47 – 0.6 m<br>• Natural frequency gait = F1 ≈ 2.0 – 2.6 | • If we take walking frequency Fwalk ≈ 2.3 which is a step mid-way between 2.0 – 2.6<br>• Fundamental frequency of LoCaDe-Exo = 14.2 Hz<br>• Separation factor = S = Fdevice/Fwalk ≈ 6.2 |